ModelIt!

Student Activity Pack

Why Your Hot Cocoa Betrays You

The Science of Thermal Energy Transfer

Grade 6 | NGSS MS-PS3-3, MS-PS3-4 | Energy and Matter: Flows, Cycles, and Conservation

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why does a metal spoon feel colder than a wooden spoon at the same temperature?

2 What happens to the temperature of hot soup if you leave it on the counter?

3 Why do you think we wear coats in winter? Do coats create heat?

4 Draw or describe how you think heat moves from a hot drink to the air around it.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Temperature Difference | Insulation Level | Container Surface Area | Air Movement

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Open Mug

No insulation, normal air movement

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Travel Mug

Insulation at 80%, normal conditions

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Windy Day

Maximum air movement, no insulation

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Radiation Rate — Heat loss through infrared energy that radiates from the hot surface to the surroundings

Container Material — Different materials conduct heat at different rates — metal vs. ceramic vs. foam

Liquid Volume — More liquid stores more thermal energy, taking longer to cool down

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

No matter what you do, your hot cocoa always gets cold — where does the heat actually go? Use evidence from your simulation to support your answers.

1 Why does blowing on hot soup cool it down faster?

2 If insulation doesn't create heat, how does a winter coat keep you warm?

3 Why is a thermos designed with a vacuum layer? What does the vacuum prevent?

4 Would your hot cocoa cool faster in a metal cup or a foam cup? Explain why.

5 How could you use thermal energy transfer knowledge to keep an ice cream from melting?

8 STEM Challenge: Design the Ultimate Hot Cocoa Keeper

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YOUR ENGINEERING MISSION

Design and test a container that keeps hot cocoa above 50°C for as long as possible using your understanding of thermal energy transfer.

THE SCENARIO

A beverage company is holding a design competition. They need a new travel mug that keeps drinks hot for at least 2 hours without electricity. Your engineering team must use model evidence to design, build, and test a prototype.

Engineering Design Thinking

1. WHAT INSULATING MATERIALS WILL YOU USE AND WHY?
2. HOW WILL YOU MINIMIZE THE SURFACE AREA EXPOSED TO AIR?
3. WILL YOUR DESIGN INCLUDE A LID? WHY IS THE TOP SURFACE IMPORTANT?
4. HOW WILL YOU MEASURE AND COMPARE YOUR DESIGN'S PERFORMANCE TO A CONTROL MUG?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Thermal Engineers and Materials Scientists design insulation systems for buildings, spacecraft, and products to control heat flow. They earn $70,000–$120,000/year.